

Maintenance optimization with imperfect maintenance of a Gamma Process

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Abstract

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1. Introduction

2. Degradation process and maintenance model

2.1. Degradation process

We assume that the deterioration process is a Gamma process, $\{X_t, t \geq 0\}$,
5 with shape parameter $a(t) = \alpha t^\beta$ and scale parameter b . The gamma process
 $\{X_t, t \geq 0\}$ is a continuous-time stochastic process such that:

- $X(0) = 0$ with probability 1
- $X(t) - X(s) \sim G(a(t) - a(s), b)$ for all $t > s \geq 0$, with density function

$\forall x \in \mathbb{R}^+$:

$$f(x) = \frac{b^{a(t)-a(s)}}{\Gamma(a(t) - a(s))} x^{a(t)-a(s)-1} e^{-bx}$$

- $X(t)$ has independent increments

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Let $\{Y_t, t \geq 0\}$ denotes the stochastic process of the maintained system.

10 We consider that the system is inspected at time $\tau, 2\tau, \dots$. At each inspection ($t_j = j\tau$), a decision is taken in respect to the level of degradation:

- if $Y_{t_j} \geq M$ then a Corrective Maintenance (CM) is performed and the system is replaced by a new one (As Good As New);
- if $L \leq Y_{t_j} < M$ then a Preventive Maintenance (PM) is performed. We
15 consider here an imperfect PM, and more precisely, an ARD_1 model, parameterized by ρ , i.e. the degradation level just after the inspection (at time t_j) is reduced by a proportional quantity of the degradation level just before the inspection: $Y_{t_j+} = (1 - \rho)Y_{t_j-}$;
- if $Y_{t_j} < L$ no action is performed,

20 where L and M are fixed thresholds ($0 < L < M$).

Let $U_i = \mathbb{1}_{L \leq Y_{i\tau} < M}$ and

Between the inspection at time $(i-1)\tau$ and the inspection at time $i\tau$, the system is assumed to evolve according to i.i.d. copies $(X_t^{(i)})_{t \geq 0}$ of the gamma process $(X_t)_{t \geq 0}$ and we denote $W_i = X_{i\tau}^{(i)} - X_{(i-1)\tau}^{(i)}$ its increment between these
25 two inspections.

Thus, before the first inspection, we have

$$\forall t \in [0, \tau), Y_t = X_t^{(1)}$$

At the first inspection, we could have:

- if $Y_{\tau-} < L$, no action is performed and then $Y_{\tau} = Y_{\tau-} = X_{\tau-}^{(1)}$.
- 30 • if $L \leq Y_{\tau-} < M$, an imperfect PM (ARD_1) is performed and then the degradation level is reduced of $\rho X_{\tau}^{(1)}$ and thus $Y_{\tau} = (1 - \rho)X_{\tau-}^{(1)}$.
- if $Y_{\tau-} > M$ then the system is replaced by a new one.

After the first inspection without replacement, we have $\forall t \in [\tau, 2\tau)$,

$$Y_t = Y_{\tau} + X_t^{(2)} - X_{\tau}^{(2)} = (1 - \rho)X_{\tau}^{(1)} + X_t^{(2)} - X_{\tau}^{(2)}$$

At the second inspection at time 2τ , we have

- if $Y_{2\tau^-} < L$, no action is performed, and $Y_{2\tau} = Y_{2\tau^-} = (1 - \rho)^{U_1} X_\tau^{(1)} + X_{2\tau}^{(2)} - X_\tau^{(2)}$.
- if $L \leq Y_{2\tau^-} < M$, an imperfect PM (ARD_1) is performed and then the degradation level is reduced of $\rho(X_{2\tau}^{(2)} - X_\tau^{(2)})$ and thus

$$Y_{2\tau} = Y_{2\tau^-} - \rho(X_{2\tau}^{(2)} - X_\tau^{(2)}) = (1 - \rho)^{U_1} X_\tau^{(1)} + (1 - \rho)(X_{2\tau}^{(2)} - X_\tau^{(2)})$$

- if $Y_{2\tau^-} > M$ then the system is replaced.

By induction, we can conclude that after i inspections without replacement, we have $\forall t \in [i\tau, (i+1)\tau)$

$$Y_t = Y_{i\tau} + (X_t^{(i+1)} - X_{i\tau}^{(i+1)})$$

and if we denote $i = \lfloor t/\tau \rfloor$, then we get

$$Y_t = \sum_{j=1}^i (1 - \rho)^{1-U_j} (X_{j\tau}^{(j)} - X_{(j-1)\tau}^{(j)}) + X_t^{(i+1)} - X_{i\tau}^{(i+1)}$$

Finally, we obtain

$$\begin{aligned} Y_{(i+1)\tau} &= Y_{i\tau} + (1 - \rho)^{U_{i+1}} (X_{(i+1)\tau}^{(i+1)} - X_{i\tau}^{(i+1)}) \\ &= \sum_{j=1}^{i+1} (1 - \rho)^{U_j} (X_{j\tau}^{(j)} - X_{(j-1)\tau}^{(j)}) \end{aligned} \quad (1)$$

2.2. Maintenance Optimization

Let C_I , C_P and C_C denote respectively the cost of an Inspection, a Preventive Maintenance and a Corrective Maintenance. As the corrective maintenances are perfect (replacement), we can compute the total cost on a renewal cycle $[0, T]$:

$$C(T) = C_C + [T/\tau]C_I + N_p C_p = C_C + [T/\tau]C_I + \sum_{i=1}^{T/\tau} \mathbb{1}_{L \leq Y_{i\tau} < M} C_p$$

and the expectation of the length of a renewal cycle is:

$$E(T) = \int P(Y_t > M) dt$$

$$E(C(T))/E(T)$$

$$E(C(T)) = C_c + E[T/\tau]C_i + E \sum_{i=1}^{T/\tau} P(M \leq Y_i\tau < L)C_P$$

$$E(T) = \int P(Y_t > L)dt$$

$$F_T(t) = P(T = \inf_{t > 0, Y_t > L} Y_t = \sum_{i < t} G_i - \sum_{j < t/\tau} Z_j Y_{j\tau}^- = G_{i-1} + Y_{(i-1)\tau}^+ \\ Y_{i\tau}^+ = (\rho)^{I_{M \leq Y_i\tau < L}} Y_{i\tau}^-$$

3. Estimation aux instants

Normalement on observe les $Y_{t_1}^-, Y_{t_2}^-, Y_{t_3}^-, \dots, Y_{t_n}^-$ Le nombre d'instant d'inspection

45 Deux possibilités: a) Des données aux instants d'inspections (savoir ou non si maintenance ou pas) b) Des données aux instants de maintenance préventive.

Si ρ aléatoire il faut estimer les paramètres de la loi de ρ

Choix d'observations a) $Y_{t_1}^-, Y_{t_2}^-, Y_{t_3}^-, \dots, Y_{t_n}^-, Y_{t_1}^+, Y_{t_2}^+, Y_{t_3}^+, \dots, Y_{t_n}^+$ On aura donc des observations de ρ en utilisant la fraction et les increments $Y_{t_i}^- - Y_{t_{i-1}}^+$

50 permettent d'estimer les paramètres du processus gamma

b) $Y_{t_1}^-, Y_{t_2}^-, Y_{t_3}^-, \dots, Y_{t_n}^-$ Plus difficile pour estimer les paramètres de la loi de ρ